
ARIMA, VAR & Multivariate Models

ECON 8310: Business Forecasting — Lecture 2

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August 24, 2026

Lecture Outline

- 1 Stationarity and Differencing
- 2 ARIMA Models
- 3 VAR Models
- 4 Key Takeaways and Roadmap

Stationarity and Differencing

Before fitting any model, the series must have a stable mean and variance.

What Is Stationarity?

Weak (Covariance) Stationarity

A time series $\{y_t\}$ is **weakly stationary** if:

1. $\mathbb{E}[y_t] = \mu$ (constant mean over time)
2. $\text{Var}(y_t) = \sigma^2 < \infty$ (constant, bounded variance)
3. $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ depends only on lag k , not on t

Why the definition earns its keep: if the mean drifts, there is no single value to forecast. If the variance grows, prediction intervals widen without bound. Stationarity is what makes the past informative about the future — it is the assumption every model in this lecture rests on.

The usual culprits are trending series — GDP, prices, sales — and variance that grows with the level. First-differencing removes a stochastic trend; a log transform stabilizes variance.

Rule of thumb: visible trend or growing variance $\Rightarrow$ likely non-stationary. Test formally with ADF or KPSS.

Unit Roots and Differencing

AR(1): $y_t = \phi_1 y_{t-1} + \varepsilon_t$. Everything turns on whether ϕ_1 is below one or exactly one.

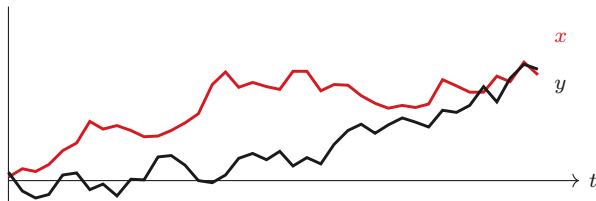
Condition	Behavior	Implication
$ \phi_1 < 1$	Shock decays, reverts to mean	Stationary AR(1)
$\phi_1 = 1$	Shock persists <i>permanently</i>	Random walk (unit root)

First differencing removes a stochastic trend: $\Delta y_t = y_t - y_{t-1}$. So $d = 0$ means already stationary, $d = 1$ means one first difference, and $D = 1$ means a seasonal difference, $\Delta_{12} y_t = y_t - y_{t-12}$.

Testing: ADF (H_0 : unit root) and **KPSS** (H_0 : stationary) test *opposite* nulls — so run both, and be suspicious when they disagree. **Do not over-difference.** A second difference on an already-stationary series induces negative autocorrelation that you then have to model away — you have manufactured the problem you are now solving. Test *after* differencing, and stop at $d = 1$ once the result is stationary.

Why the Rule Exists: Spurious Regression

Differencing is not bookkeeping. Skip it and the statistics actively lie to you.



Two independent random walks — **no shared cause, no relationship at all**. Their *changes* correlate at -0.13, which is the truth. But their *levels* correlate at +0.62, and that is what OLS sees.

Granger and Newbold (1974) showed this rejects “no relationship” about **three-quarters** of the time at the 5% level. The regression is not finding a relationship — it is finding that both series wander. **Difference first.**

ACF and PACF: Model Identification

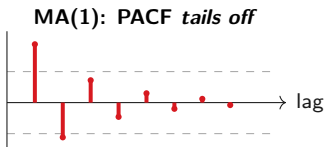
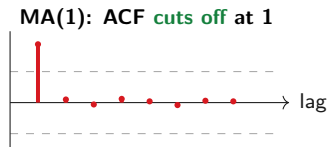
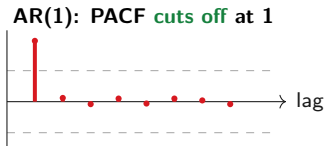
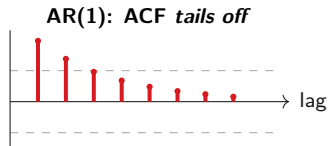
The **autocorrelation function**, $\rho_k = \text{Corr}(y_t, y_{t-k})$, measures how y_t relates to its own past. The **partial** autocorrelation measures the same thing while controlling for the lags in between — which is what separates a direct effect from one inherited through intermediate lags.

Read them together and they identify the model:

Model	ACF	PACF
AR(p)	Tails off geometrically	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off geometrically
ARMA	Tails off	Tails off
RW / I(1)	Decays slowly	Big spike at lag 1

Always unit-root test and difference *before* reading either plot. The ACF of a non-stationary series decays slowly no matter what the underlying process is, so it tells you nothing about p or q — only that you have not differenced yet.

What Those Patterns Actually Look Like



Read the pair, never one alone. The **sharp cut-off** tells you the order; the *slow decay* tells you which side it belongs on. Dashed lines are the significance bands — spikes inside them are noise, not signal.

ARIMA Models

Combine differencing with AR and MA components into one unified model.

ARIMA(p,d,q)

ARIMA(p,d,q) Model

Apply an ARMA(p,q) to the d -times differenced series: $y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$

p is the number of AR lags, d the number of differences, q the number of MA lags. Several familiar models are just corners of this space: **ARIMA(0,1,0)** is a random walk, **ARIMA(1,0,0)** is an AR(1), and **ARIMA(0,1,1)** is algebraically equivalent to simple exponential smoothing from Lecture 1.

A forecast by hand. Take ARIMA(1,1,0) with $\hat{\phi}_1 = 0.5$. If $y_{100} = 120$ and $y_{99} = 118$, then $\Delta y_{100} = 2$, and

$$\hat{y}_{101|100} = 120 + 0.5 \times 2 = \mathbf{121}.$$

In Python: `pmdarima.auto_arima()` searches orders automatically; `statsmodels.SARIMAX` fits any order you specify.

SARIMA: Adding Seasonal Terms

For monthly or quarterly data, autocorrelation often appears at lags $m, 2m, 3m, \dots$

$\text{SARIMA}(p,d,q)(P,D,Q)[m]$ adds seasonal AR and MA polynomials alongside the non-seasonal ones.

The useful mental model is **two ARIMA models running at different time scales**: the non-seasonal part (p, d, q) captures week-to-week dynamics, while the seasonal part $(P, D, Q)[m]$ captures the year-over-year pattern at lag m .

Practical workflow:

1. ADF / KPSS $\rightarrow$ choose d
2. Seasonal unit root test $\rightarrow$ choose D
3. ACF / PACF on the differenced series $\rightarrow$ guide p, q, P, Q
4. Minimize AIC, then check the residual ACF with a Ljung-Box test

For monthly retail data with trend and December seasonality, $\text{SARIMA}(1,1,1)(0,1,1)[12]$ is a strong starting point — and a good default to beat before trying anything more elaborate.

VAR Models

When two series influence each other, model them together symmetrically.

Why Use Multiple Series?

Every model so far — ETS, ARIMA — uses only y_t 's own history. But business variables are linked: consumer sentiment today moves retail spending next month, interest rates move housing starts within two quarters, advertising spend moves sales over the following weeks.

The principle underneath all of it: **if x_t carries information about y_{t+h} that is not already in y_t 's own history, including x_t reduces forecast error.** Granger causality (Granger 1969) is the formal test of exactly that claim, and we get to it in two slides.

That leaves a modelling choice, and it turns on whether the influence runs one way or both:

VAR — symmetric

- Every variable predicts every other
- Use when influence runs both ways
- No variable is privileged

ARIMAX — one direction

- x_t drives y_t , not the reverse
- Use when the causal story is clear
- Needs a forecast of x_t itself

The VAR(p) Model

For two stationary series y_{1t} and y_{2t} , each equation regresses on the lags of *both*:

$$\begin{cases} y_{1t} = c_1 + a_{11} y_{1,t-1} + a_{12} y_{2,t-1} + \varepsilon_{1t} \\ y_{2t} = c_2 + a_{21} y_{1,t-1} + a_{22} y_{2,t-1} + \varepsilon_{2t} \end{cases}$$

The cross terms are where the value is: a_{12} is how much $y_{2,t-1}$ predicts y_{1t} , which is precisely what a univariate AR throws away. If $a_{12} = a_{21} = 0$, the system collapses back into two separate AR models and you have learned that too. Each equation is estimated by OLS; the order p is chosen by BIC.

Parameter explosion. A VAR(p) with k variables has $k^2 p$ coefficients. At $k = 5$ and $p = 4$ that is **100 coefficients** — typically more than a monthly business series has observations. Keep $k \leq 4$ unless you are using a LASSO-VAR.

Python: `statsmodels.tsa.api.VAR; var.fit(maxlags=p, ic='bic')` selects the order.

Granger Causality Test

Granger Causality

x_t **Granger-causes** y_t if past values of x_t help predict y_t *beyond what y_t 's own past already explains.*

The test: fit the y -equation with and without the x -lags, then run an F -test of H_0 : all x -lag coefficients are zero. Rejecting means x_t Granger-causes y_t .

A concrete question. Does consumer sentiment (UMCSENT) Granger-cause retail sales (RSXFS)? The null is that every sentiment lag is zero in the retail equation. We will run exactly this in the lab.

Granger causality is not structural causality. It tests *predictive content*, not whether x mechanically brings y about — a rooster reliably predicts sunrise. Report it as “predictive of,” never “causes.”

Impulse response functions trace how a shock propagates through the system over time.

ARIMAX: Exogenous Variables with ARIMA Errors

When one variable clearly *drives* another rather than the relationship being symmetric, use ARIMAX.

ARIMAX Model

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \eta_t, \quad \eta_t \sim \text{ARIMA}(p, d, q)$$

$\mathbf{x}_t$ are exogenous regressors; η_t absorbs whatever autocorrelation is left in the residuals.

Use it when the causal direction is clear (advertising $\rightarrow$ sales), you already have x_t or can forecast it, and adding x_t actually reduces cross-validated forecast error — check that last one rather than assuming it.

The catch that surprises people: to forecast y_{t+h} you need x_{t+h} — a forecast of your own predictor. If that is unavailable, ARIMAX buys you nothing at forecast time and you should use a VAR instead, which generates its own.

Python: `SARIMAX(y, exog=X, order=(p,d,q)).fit()`

Key Takeaways and Roadmap

Stationarity, ARIMA, VAR, and ARIMAX form the classical time series toolkit.

Key Takeaways

1. **Stationarity** is required before fitting time series models. Test with ADF (H_0 : unit root) and KPSS (H_0 : stationary). First-differencing removes a stochastic trend.
2. **ACF** identifies MA order; **PACF** identifies AR order. Always inspect after differencing, not before.
3. **ARIMA**(p, d, q) combines differencing with AR/MA. SARIMA adds seasonal polynomials for periodic autocorrelation. Use `auto_arima` for automatic selection.
4. **VAR** models all variables symmetrically and captures cross-variable dynamics. Keep $k \leq 4$. Use **Granger causality** to test predictive content.
5. **ARIMAX** adds exogenous predictors to ARIMA. Use when causal direction is clear and x_t is forecastable.

Next: Generalized Additive Models (GAMs) — flexible, interpretable nonlinear forecasting with Prophet and pyGAM.

References I

-  Granger, C. W. J. (1969). "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods". In: *Econometrica* 37.3, pp. 424–438.
-  Granger, C. W. J. and Paul Newbold (1974). "Spurious Regressions in Econometrics". In: *Journal of Econometrics* 2.2, pp. 111–120.